

Neural Network Aided Impulsive Perturbation Decoding for Short Raptor-Like LDPC Codes

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Abstract—As short packet communications become more important, multi-round belief propagation decoding with impulsive perturbation (MBPD-IP) was proposed to improve the decoding of short low-density parity-check (LDPC) codes. To operate the MBPD-IP effectively, it is crucial to select proper symbols to be perturbed in as few trials as possible. Unfortunately, existing perturbation symbol selection criteria are heuristic and not effective for special codes such as raptor-like (RL) LDPC codes. In this letter, a neural network (NN) based perturbation symbol selection scheme for MBPD-IP is proposed where the symbols to be perturbed are selected from a pre-trained NN. It is shown that the proposed scheme performs significantly better than existing schemes for the RL LDPC code of 5G new radio and still works well for plain code structures.

Index Terms—Low density parity check codes, perturbation decoding, deep learning, neural network, short packet communication.

I. INTRODUCTION

AS ULTRA reliable and low latency communication (URLLC) services emerge, short packet communications has drawn more attention [1], [2]. Because the capacity is limited in communication in finite block length [3] and the latency requirement is tight, short codes were studied in a wider spectrum [4] and the demand on maximum-likelihood-like (ML-like) decoding algorithm is increasing.

In 1995, a soft decision decoding based on ordered statistics for linear block codes [5] that can approach ML performance was proposed. However, the complexity increases quickly as the search space grows. A linear programming based decoding was proposed for linear codes [6] and the alternating direction method of multipliers (ADMM) was used for linear programming decoding of low density parity check (LDPC) codes [7].

To approach the ML performance in low complexity or to overcome the intrinsic weaknesses of decoding algorithms, learning aided decoding have been proposed [8]–[15]. An end to end supervised learning was used to train a neural-network (NN) decoder [8], although training was effective only for very

short codes. A deep learning-aided ADMM penalized decoding was applied to LDPC codes [10]. Also, hard decision based bit-flipping decoding was enhanced by reinforcement learning for the decoding of linear codes [11]. A weighted belief propagation (BP) decoding was optimized by a deep learning technique, where variable and check nodes are used as neurons that form the corresponding NN [12], [13].

Another approach toward ML performance is based on repetitive post processing. To overcome the finite length problems of short LDPC codes, multi-round BP decoding with perturbation (MBPD-P) has been proposed [16]–[20]. Perturbation is a process of making a small intentional change in the received signal, and the MBPD-P schemes conduct the conventional BP decoding iteratively with such input perturbation until a valid codeword is found. According to the reported performance for the CCSDS (128, 64) LDPC code, the MBPD-P's performance is better than or on par with the deep learning-aided ADMM penalized decoder [10] and the pruned neural BP decoder [15]. Therefore, improving MBPD-P in terms of performance and complexity is meaningful. In [21], Lee *et al.* proposed the multi-round BP decoding with impulsive perturbation (MBPD-IP) which is based on the special perturbation that applies to only a few symbols. The MBPD-IP outperforms conventional MBPD-P schemes under the constraint on the number of perturbation rounds.

While the performance of the MBPD-P is improved as the number of perturbation rounds increases, the overall decoding complexity and maximum decoding latency increase. Hence, it is desired to minimize the number of perturbation rounds by effectively selecting symbols to be perturbed. However, as the decoding result is dependent on the specific realization of code construction and the channel noise realization, it is hard to find optimal criteria for perturbation symbol selection. Thus, the existing selection criteria have been heuristically designed in all MBPD-P algorithms [16]–[21].

As an example, the number of sign mismatches of variable nodes (nSMV) was proposed for the perturbation symbol selection criterion in [21]. The nSMV keeps the signs of variable node messages during BP decoding to assess the symbol reliability. It was shown that among the existing reliability measures, the nSMV shows the best decoding performance for ordinary regular and irregular LDPC codes [21]. However, for some special code structures, such as Raptor-like (RL) LDPC codes of 5G new radio [22], some codeword symbols with low reliability cannot be effectively detected by the nSMV criterion. The RL LDPC codes have degree 1 variable nodes for good low rate performance and rate compatibility. Since the messages of the degree 1 variable nodes do not update during

Manuscript received August 24, 2021; revised October 24, 2021; accepted October 27, 2021. Date of publication November 4, 2021; date of current version February 17, 2022. This work was supported by Samsung Research, Samsung Electronics Co., Ltd. The associate editor coordinating the review of this article and approving it for publication was J. Hu. (*Corresponding author: Sang-Hyo Kim.*)

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Digital Object Identifier 10.1109/LWC.2021.3125578

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the BP decoding process, and consequently, the reliability of such nodes cannot be measured by nSMV. In order to effectively operate the MBPD-IP regardless of the code structure, a more general reliability measure is required.

In this letter, we propose a new perturbation symbol selection scheme aided by an NN which is adaptive to the code structure and specially effective to RL LDPC codes. Owing to the simple implementation and good performance, we only consider the impulsive perturbation (IP). The NN learns through the symbol indices with successful IP decoding. In addition, several perturbation symbol patterns are effectively learned by applying the weighted loss. We compare the error performance and average number of perturbation rounds of the proposed scheme with conventional schemes [21]. Numerical results show that the performance of the proposed symbol selection is better than the existing criteria for the short RL LDPC codes. Moreover, the average decoding complexity is lower for the proposed scheme.

The remainder of this letter is organized as follows. In Section II, we briefly introduce the MBPD-IP schemes. Then the proposed perturbation symbol selection scheme is presented in Section III. The performance and complexity of the proposed scheme are evaluated in Section IV. Section V concludes this letter.

II. MULTI-ROUND BELIEF PROPAGATION DECODING WITH IMPULSIVE PERTURBATION

In this section, we introduce the basic system model and the MBPD-IP which is a good low-complexity MBPD-P.

A. System Model

The system model is depicted in Fig. 1. Let $\mathbf{C} = \{\mathbf{c} | \mathbf{c}\mathbf{H}^T = \mathbf{0}\}$ be an (N, K) binary LDPC code where $\mathbf{H}$ is a sparse $(N - K) \times N$ parity check matrix and $\mathbf{c} = (c_0, \dots, c_{N-1}) \in \{0, 1\}^N$ is a codeword. The LDPC code is concatenated with cyclic redundancy check (CRC) code for error detection. We consider the binary phase shift keying (BPSK) modulated signal $\mathbf{x} = (x_0, \dots, x_{N-1})$, and $x_i = (-1)^{c_i}$ corresponding to the codeword. We assume an additive white Gaussian channel (AWGN) such that $\mathbf{r} = (r_0, \dots, r_{N-1}) = \mathbf{x} + \mathbf{z}$ where $\mathbf{z} = (z_0, \dots, z_{N-1})$ and z_i 's are independent Gaussian random variables with zero mean and variance σ^2 . Let $\mathbf{l} = (l_0, \dots, l_{N-1})$ be a vector of the intrinsic log-likelihood ratio (LLR) of $\mathbf{c}$ for the observation $\mathbf{r}$. If CRC check is satisfied, the decoder outputs a codeword estimate $\hat{\mathbf{c}} = (\hat{c}_0, \dots, \hat{c}_{N-1})$. Otherwise, the received signal is perturbed and the decoding process is performed until a CRC-pass or the number of perturbation rounds reaches the maximum value, T .

B. MBPD-IP

Perturbation is a process of making a small intentional change in the received signal, and the MBPD-P schemes conduct the conventional BP decoding iteratively with such input perturbation until a valid codeword is found [21]. While performing the MBPD-P, in order to decrease the decoding complexity, it is desired to find a successful perturbation pattern early. To achieve this goal, a concentration of perturbation

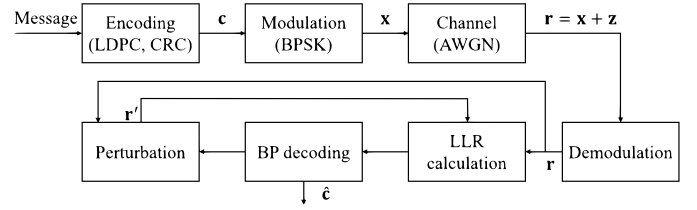


Fig. 1. Block diagram of MBPD-IP.

to one or a few symbols was proposed in pair with an efficient perturbation symbol selection criterion [21]. The proposed scheme is referred to as the MBPD-IP where IP is a modification of the received signal by adding impulsive noise of opposite sign on one or a few symbols. As a result, decoding performance and complexity were improved. Also, the proposed scheme in [21] supports a parallel implementation of MBPD-IP.

The process of MBPD-IP is discussed in more detail. The first step of perturbation is to select the space of perturbation from the received signal. The perturbation space $\mathcal{R} = \{\pi(0), \dots, \pi(\tau - 1)\}$ is the set of the indices of code-word symbols with low reliability, where τ is the perturbation space size and $\pi(i) \in \{0, \dots, N - 1\}$ is the index permutation. In [21], nSMV was proposed for the perturbation symbol selection criterion. The nSMV is defined by the number of mismatches between the extrinsic LLR and the posterior LLR over the initial BP decoding, which is regarded as a measure of symbol reliability.

Let $l_E^i(j)$ be the extrinsic LLR and $l^i(j)$ be the posterior LLR of the j -th variable node at the i -th iteration of component BP decoding. Then the nSMV for the j -th variable node is defined as

$$m_j = \left| \{i | l_E^i(j) l^i(j) < 0, 1 \leq i \leq I_{\max}\} \right|,$$

where $I_{\max}$ is the maximum number of BP decoding iterations. Symbols with larger nSMV are considered less reliable. If there are more than one variable node with the maximum nSMV, then the magnitude of intrinsic LLR is used as the secondary rule.

After that, a set of T perturbation patterns, $\mathcal{P} = \{\mathbf{p}^0, \dots, \mathbf{p}^{T-1}\}$, is generated from the selected perturbation symbol, where $\mathcal{P}$ is an ordered set of binary τ -tuples, $\mathbf{p}^t = (p_0^t, \dots, p_{\tau-1}^t) \in \{0, 1\}^\tau$. The IP applies to the received signal corresponding to the perturbation pattern as

$$r'_{\pi(i)} = \begin{cases} -\text{sgn}(r_{\pi(i)})A, & \text{if } p_i^t = 1, \\ r_{\pi(i)}, & \text{if } p_i^t = 0, \end{cases}$$

where $0 \leq i \leq \tau - 1$, A is a large value, and $\pi(i)$ is the index order of symbol reliability of $\mathbf{r}$.

When we apply the perturbation patterns, $\mathbf{p}$'s with lower weight are chosen first. Among $\mathbf{p}$'s of the same weight, a lexicographical ordering is applied. The final step of perturbation is to input the perturbed signal $\mathbf{r}' = (r'_0, \dots, r'_{N-1})$ to the BP decoder. The LLRs of perturbed signal $\mathbf{l}' = (l'_0, \dots, l'_{N-1})$ for the following BP decoding is set as $\mathbf{l}' = 2\mathbf{r}'/\sigma^2$. The perturbation operations are repeated up to T rounds until a valid codeword is founded.

III. NN-AIDED IMPULSIVE PERTURBATION SYMBOL SELECTION

It was observed that the existing perturbation symbol selection criteria, such as nSMV, are not effective to the special structure of RL LDPC codes having many degree 1 nodes. In this section, we propose an NN-aided perturbation symbol search algorithm which can reflect the symbol reliability more adaptively to the code structure.

A. Framework of NN-Aided MBPD-IP

In this section, we propose a perturbation space selection method based on an NN for effective MBPD-IP. First, the NN-based MBPD-IP framework is briefly introduced. For each received vector $\mathbf{r}$, an NN is used to predict the perturbation patterns that are most likely to lead to a decoding success. For simple and effective search, only one-bit perturbation is considered. Multi-bit perturbation may be considered in a future study. The NN takes in the received vector, $\mathbf{r}$ as the input and outputs $\hat{\mathbf{b}} = (\hat{b}_0, \dots, \hat{b}_{N-1})$, where $\hat{b}_i$ is the estimated probability that decoding will succeed if perturbation applies to the received symbol r_i . The decoder proceeds the impulsive perturbation to one-bit at a time and the one-bit perturbation is chosen by the order of decreasing chance of successful decoding with respect to $\hat{\mathbf{b}}$. The number of one-bit perturbation patterns that are likely to lead to a successful decoding depends on $\mathbf{r}$.

More specifically, the NN is used to produce $\mathcal{R}$ of the MBPD-IP in the previous section. Let $\pi(i)$ be the permutation that indicates the index of the i -th highest element of the NN output $\mathbf{b}$. The indices of T highest values are selected as the perturbation symbols for one-bit IP;

$$\mathcal{R} = \{\pi(i) | 0 \leq i \leq T-1\},$$

where $\hat{b}_{\pi(i)} \geq \hat{b}_{\pi(i+1)}$. Finally, MBPD with one-bit IP is performed up to T times where the perturbation pattern is given as a weight one binary vector with a single one at the $\pi(i)$ -th entry for $\pi(i) \in \mathcal{R}$.

To train the NN, data can be obtained from the simulation results of running MBPD-IP with full IP pattern search. All one-bit perturbations are tested and the binary vector $\mathbf{b}$ that indicates the successful one-bit perturbations by setting ones at the bit locations becomes the label. Thus, the Hamming weight of $\mathbf{b}$ is the number of successful IPs.

B. Training Data Generation

In this subsection, the data generation method for the NN-aided perturbation symbol selector is explained. The training data set is a set of data points, each of which consists of an input and its label pair, denoted by

$$\mathcal{D} = \{(\mathbf{r}^j, \mathbf{b}^j) | 0 \leq j \leq M-1\}.$$

In the proposed method, the label and the binary vector of which ones indicate the positions of the successful one-bit perturbation, respectively. The training data generation method is described in Algorithm 1. The algorithm runs until we get M data points, i.e., $|\mathcal{D}| = M$ where $|\cdot|$ denotes the set cardinality. For every outer iteration, a received signal $\mathbf{r}$ is generated with randomly generated codeword and channel noise

Algorithm 1 Data Generation

Input: $M, E_b/N_0, N, \mathcal{P}_{\text{one}}, I_{\text{max}}, \mathcal{D} \leftarrow \emptyset$

- 1: **while** $|\mathcal{D}| < M$ **do**
- 2: Generate a received signal $\mathbf{r}$ at given E_b/N_0 , $\mathbf{b} \leftarrow \mathbf{0}_N$
- 3: **for** $t = 0 : N-1$ **do**
- 4: Perform one-bit IP $\mathbf{r}'$ with $\mathbf{p}^t$
- 5: Calculate the LLRs of $\mathbf{r}'$ and perform a BP decoding with the maximum iteration of I_{max}
- 6: **if** LDPC syndrome and CRC check are satisfied **then**
- 7: $b_t \leftarrow 1$
- 8: **end if**
- 9: **end for**
- 10: **if** $w_H(\mathbf{b}) \geq 1$ **then**
- 11: $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{r}, \mathbf{b})\}$
- 12: **end if**
- 13: **end while**

at given E_b/N_0 , and all elements of the temporal label is initialized as zero, where $\mathbf{0}_N$ is the zero vector of length N . Then for every inner iteration, the received signal is perturbed with a one-bit perturbation pattern $\mathbf{p}^t$ and the BP decoding is performed with maximally I_{max} iterations. Here, the set of all one-bit perturbation patterns is $\mathcal{P}_{\text{one}} = \{\mathbf{p}^0, \dots, \mathbf{p}^{N-1}\}$, where $\mathbf{p}^i = (p_0^i, \dots, p_{N-1}^i) \in \{0, 1\}^N$ and

$$p_j^i = \begin{cases} 1, & \text{if } j = i, \\ 0, & \text{if } j \neq i. \end{cases}$$

Decoding succeeds if both LDPC syndrome check and CRC check are satisfied. Then, the elements of label corresponding to the successful one-bit perturbation pattern is replaced by 1. For example, let us assume a received signal of $N = 10$ for which the generic BP decoding fails. If the IP decoding succeeds in the 4th, 5th, and 8th symbol perturbation, respectively, the label of this signal is $\mathbf{b} = (0, 0, 0, 1, 1, 0, 0, 1, 0, 0)$. If at least a one-bit perturbation pattern has lead to successful decoding, then the Hamming weight of the label, $w_H(\mathbf{b})$ is nonzero. For the successful case, the received signal and label pair, $(\mathbf{r}, \mathbf{b})$ is added to the temporary data set, $\mathcal{D}$.

C. Training of MLP

The goal of training is an optimization of the weights and biases for minimum loss between output $\hat{\mathbf{b}}$ and label $\mathbf{b}$. In this letter, we utilized an MLP, which is a fully-connected feed-forward NN model. The input and output layers has N nodes where $\mathbf{r}$ is taken as the input. There are two hidden layers of size $2N$ and N nodes, respectively. Each output node is corresponding to the variable node of the same index in the LDPC graph. The rectified linear unit and a linear function is used as the activation function in the hidden layers and the output layer, respectively.

For the training of the IP space selector, we train the NN with the data generated by Algorithm 1. The NN is optimized via supervised learning. Let $\text{MLP}_\theta(\mathbf{r}, \mathbf{b})$ be an MLP model with parameter set θ . As the desired function of $\text{MLP}_\theta(\mathbf{r}, \mathbf{b})$ is the perturbation symbol selector, the output $\hat{\mathbf{b}}$ of the MLP indicates the likelihood of each element for one-bit IP success.

TABLE I
HYPER-PARAMETERS FOR NN TRAINING

Size of training data per E_b/N_0	250,000
Training E_b/N_0 [dB]	(1, 2, 3, 4)
Size of test data per E_b/N_0	10,000
Test E_b/N_0 [dB]	(2, 2.5, ... 5.5)
Learning rate	0.0001
Optimization method	Adam optimization
Size of mini-batch	500
The number of epoch	500

The number of ones in the label indicates the number of successful one-bit perturbation patterns. Thus, the Hamming weight of the label, $w_H(\mathbf{b})$, is an indicator showing perturbation decodability, or diversity of the corresponding input pair. If $w_H(\mathbf{b})$ is large, there is higher probability of decoding success. The reason is that if only one among $w_H(\mathbf{b})$ symbols is included in the set of indices of T highest NN symbol outputs, then the decoding will be successful. Thus, we get a diversity of $w_H(\mathbf{b})$. From the NN's perspective, the case of small decoding diversity is a challenging task whereas the case of large decoding diversity is a rather easy task with respect to successful decoding. Therefore, we intend to balance the difficulty of the tasks of different diversity. To promote the learning of such a data point with low decoding diversity, we apply a weighted loss function depending on $w_H(\mathbf{b})$.

The weighted loss, which is one of the solutions to the class imbalance problem [23], is applied the predetermined label $\mathbf{b}$ associated with $\mathbf{r}$. The scaling factor is given as $w = w_H(\mathbf{b})^{-1}$, where w is dependent on the received signal $\mathbf{r}$. The modified mean squared error (MSE) is defined as

$$L = \frac{w}{N} \sum_{i=0}^{N-1} (b_i - \hat{b}_i)^2, \quad (1)$$

where b_i and $\hat{b}_i$ are symbols of the label and the output, respectively. The cross entropy was tested for loss function, however the MSE performed better when the maximum number of perturbation rounds is more than a few.

The NNs are trained by data sequentially taken from the training data, which is generated from a high E_b/N_0 to a low E_b/N_0 . Since the NN is pre-trained and used in the perturbation decoding without further training, the complexity of such offline training is not crucial and a sufficiently large number of epochs can be used until the performance saturates.

IV. NUMERICAL RESULTS

In this section, we numerically evaluate proposed criterion for MBPD-IP. The effectiveness of the proposed criterion are shown and compared with the existing perturbation symbol selection criteria, which are the nSMV combined with intrinsic LLR as the secondary rule and the intrinsic LLR alone [21].

The decoding error performance and computational complexity of MBPD-P algorithms are of interest. For the simulation, a short RL LDPC code is used. We consider (104, 20) RL LDPC code concatenated with the CRC-8 code of a generator $g(x) = x^8 + x^4 + x^3 + x^2 + 1$ [22]. For the BP decoder, the normalized min-sum (NMS) decoder is considered with $I_{\max} = 20$ and a scaling factor of 0.8. The maximum number of perturbation rounds T is set to 10 and 40, respectively.

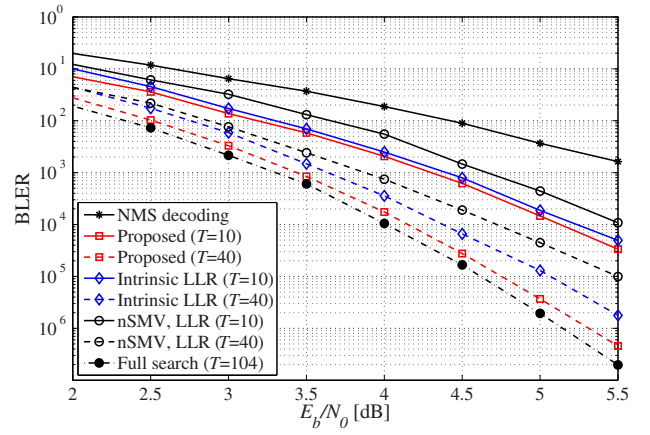


Fig. 2. Performance of MBPD-IP with the proposed NN based criterion and conventional criteria (nSMV combined with intrinsic LLR, intrinsic LLR) for the (104, 20) LDPC code with CRC-8.

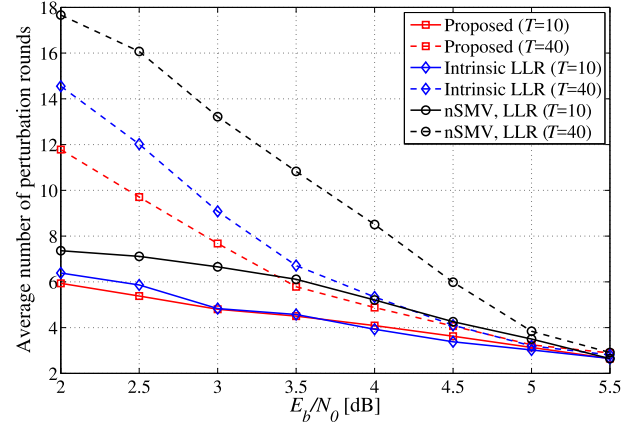


Fig. 3. Average number of perturbation rounds of MBPD-IP with the proposed scheme and conventional criteria (nSMV combined with intrinsic LLR, intrinsic LLR) for the (104, 20) LDPC code with CRC-8.

The hyper-parameters regarding the NN-model is given in Table I. The training data is generated for $E_b/N_0 = 1, 2, 3, 4$ dB and the number of training data for each E_b/N_0 is 2.5×10^5 . For training, the mini-batch size, dropout rate and learning rate are set to 1000, 0.05, and 0.0001, respectively. The number of epochs is 500. The weighted loss function in (1) is used for the loss function and the stochastic gradient descent method with ADAM optimizer [24] is used.

Fig. 2 shows the block error rate (BLER) performance of different criteria for the IP decoding. The proposed method achieves a performance gain of 0.35 dB for $T = 10$, and 0.45 dB for $T = 40$ at a BLER of 10^{-3} over the nSMV criterion combined with intrinsic LLR. The performance gain over intrinsic LLR is 0.1 dB for $T = 10$ and 0.25 dB for $T = 40$. We can confirm that the proposed scheme has better performance than the nSMV with intrinsic LLR or the intrinsic LLR.

Fig. 3 compares the average number of perturbation rounds. The proposed scheme shows 19%, and 43% reduction in average number of perturbation rounds as compared to the intrinsic LLR, and nSMV combined with intrinsic LLR for $T = 40$ at $E_b/N_0 = 3$ dB. Therefore, the proposed NN-aided perturbation symbol selection scheme performs better than the

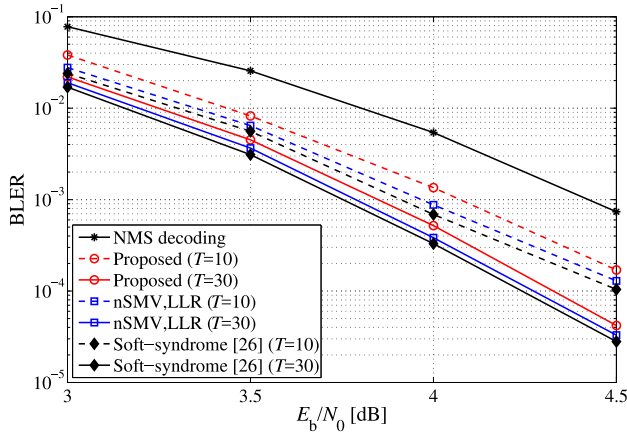


Fig. 4. Performance of MBPD-IP with the proposed NN based symbol selection and conventional criteria (nSMV combined with intrinsic LLR) for the CCSDS (128, 64) LDPC code with CRC-8.

existing schemes for both the error correcting performance and computational complexity.

Although the proposed NN-aided scheme is developed for the special structure of RL LDPC codes, this scheme adapts well to non-RL codes as the proposed scheme performs as well as the nSMV combined with intrinsic LLR for ordinary regular and irregular LDPC codes according to simulation results.

In Fig. 4, the performance of the proposed scheme is compared with the nSMV combined with intrinsic LLR for a (128, 64) CCSDS LDPC code [25]. The code is a non-Raptor-like and plain irregular LDPC code with variable node degrees of three and five and a regular CN degree of eight. The decoding methods are compared for $T = 10$ and 30. It is seen that the proposed NN-aided scheme performs robustly for such an ordinary structure of LDPC codes, although a slight performance degradation, 0.05dB at BLER 10^{-3} for $T = 30$, is observed.

It is worth noting that the degradation can be overcome by an input processing for non-Raptor-like structures. In Fig. 4, the performance of the NN-aided symbol selection with additional input processing [26] is also shown. For the NN, soft-syndrome [14] is concatenated to the input $\mathbf{r}$. The NN is trained by supervised learning as in this letter. The performance is even better by 0.03dB at BLER 10^{-3} .

V. CONCLUSION

In this letter, we proposed an NN-aided perturbation symbol selection method for MBPD-IP of short LDPC codes. In the proposed method, the NN is trained with a received signal and perturbation pattern pair. We devised weighted loss function for effective training of the NN with the training data with low perturbation decodability. Compared with existing criteria, the proposed scheme provides a better performance for RL LDPC codes which have a special structure. Also, the proposed scheme is valid for non-RL structures. As we only considered one-bit perturbation in this work, NN-aided multi-bit perturbation pattern search can be studied for a future work.

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